

Effect of high-intensity exercise training on lactate/H⁺ transport capacity in human skeletal muscle

Henriette Pilegaard, Kristian Domino, Thomas Noland, Carsten Juel, Ylva Hellsten, Andrew P. Halestrap and Jens Bangsbo

Am J Physiol Endocrinol Metab 276:E255-E261, 1999. ;

You might find this additional info useful...

This article cites 33 articles, 18 of which you can access for free at:
<http://ajpendo.physiology.org/content/276/2/E255.full#ref-list-1>

This article has been cited by 33 other HighWire-hosted articles:
<http://ajpendo.physiology.org/content/276/2/E255#cited-by>

Updated information and services including high resolution figures, can be found at:
<http://ajpendo.physiology.org/content/276/2/E255.full>

Additional material and information about *American Journal of Physiology - Endocrinology and Metabolism* can be found at:
<http://www.the-aps.org/publications/ajpendo>

This information is current as of July 29, 2016.

Effect of high-intensity exercise training on lactate/H⁺ transport capacity in human skeletal muscle

HENRIETTE PILEGAARD,¹ KRISTIAN DOMINO,¹ THOMAS NOLAND,¹
CARSTEN JUEL,¹ YLVA HELLSTEN,¹ ANDREW P. HALESTRAP,² AND JENS BANGSBO¹

¹Copenhagen Muscle Research Centre, The August Krogh Institute, University of Copenhagen, 2100 Copenhagen, Denmark; and ²Department of Biochemistry, School of Medical Sciences, University of Bristol, Bristol BS8 1 TD, United Kingdom

Pilegaard, Henriette, Kristian Domino, Thomas Noland, Carsten Juel, Ylva Hellsten, Andrew P. Halestrap, and Jens Bangsbo. Effect of high-intensity exercise training on lactate/H⁺ transport capacity in human skeletal muscle. *Am. J. Physiol.* 276 (Endocrinol. Metab. 39): E255–E261, 1999.—The present study examined the effect of high-intensity exercise training on muscle sarcolemmal lactate/H⁺ transport and the monocarboxylate transporters (MCT1 and MCT4) as well as lactate and H⁺ release during intense exercise in humans. One-legged knee-extensor exercise training was performed for 8 wk, and biopsies were obtained from untrained and trained vastus lateralis muscle. The rate of lactate/H⁺ transport determined in sarcolemmal giant vesicles was 12% higher ($P < 0.05$) in the trained than in untrained muscle ($n = 7$). The content of MCT1 and MCT4 protein was also higher (76 and 32%, respectively; $n = 4$) in trained muscle. Release of lactate and H⁺ from the quadriceps muscle at the end of intense exhaustive knee-extensor exercise was similar in the trained and untrained leg, although the estimated muscle intracellular-to-interstitial gradients of lactate and H⁺ were lower ($P < 0.05$) in the trained than in the untrained muscle. The present data show that intense exercise training can increase lactate/H⁺ transport capacity in human skeletal muscle as well as improve the ability of the muscle to release lactate and H⁺ during contractions.

monocarboxylate transporters; blood flow; pH

THE DECLINE in muscle force during development of muscle fatigue during high-intensity exercise is associated with accumulation of lactate and a concomitant lowering of pH in the muscle. Several studies have demonstrated that reduced muscle pH can interfere with the excitation-contraction coupling process (9), and high muscle lactate concentrations have also been reported to inhibit Ca²⁺ release from the sarcoplasmic reticulum (8). Therefore, the removal of lactate and H⁺ from skeletal muscle is likely to be of importance for the ability to maintain force during high-intensity exercise.

It is well known that the main part of the sarcolemmal lactate flux is mediated by the lactate/H⁺ transport systems in both animal (13, 14, 32, 35) and human (17) skeletal muscle, and the lactate/H⁺ transporters have been shown to be the muscle membrane transport systems with the highest capacity for H⁺ removal in rat muscle (15). Several monocarboxylate transporter (MCT) proteins have been identified and cloned (10–12,

31). In rat skeletal muscle, MCT1 has been found predominantly in oxidative muscles and only in small amounts in glycolytic muscles (22, 36), whereas MCT2 seems to be absent or present only in low concentrations in rat skeletal muscle (10, 12). Recently, MCT3 to MCT7 were cloned and sequenced (31), and the MCT4 isoform (formerly called MCT3) has been demonstrated in both oxidative and glycolytic rat muscle (31).

The lactate/H⁺ transport capacity of rat skeletal muscle has been reported to be enhanced by training (24, 28) as well as by chronic low-frequency stimulation (21). MCT1 was also found to be increased with chronic electrical stimulation and high-intensity training in rats (23) and after training in humans (6). Furthermore, a cross-sectional study showed that speed endurance-trained athletes have a higher capacity to transport lactate than untrained and less trained subjects (27). The latter finding suggests that lactate/H⁺ transport can be improved by training also in humans, but it can not be excluded that the higher lactate/H⁺ transport capacity observed in the athletes was due to biological selection.

Thus the purpose of the present study was to examine whether training can induce changes in muscle lactate/H⁺ transport capacity and lactate/H⁺ transporter proteins in humans and furthermore to evaluate to what extent such alterations will affect lactate and H⁺ release during intense exercise.

METHODS

Subjects

Seven male subjects, aged from 20 to 24 yr, with an average height and weight of 183 (range: 174–187) cm and 81.8 (range: 55.0–107.7) kg, respectively, participated in the study. The body mass was similar before and after the training period. The subjects were habitually physically active without taking part in competitive sports, and they maintained their normal activity pattern during the training period. The subjects were informed about the experimental procedure before they gave their informed consent, and the study was approved by the Copenhagen Ethics Committee.

The mass of the quadriceps femoris muscle of the untrained and the trained leg was estimated from anthropometric and skinfold measurements of the thigh (3), and no difference in the mass of the quadriceps muscle of the trained (3.3 ± 0.1 kg) and the untrained leg (3.3 ± 0.2 kg) was observed.

Training

The training consisted of dynamic one-legged knee extensor exercise in the supine position on a modified Krogh bicycle

The costs of publication of this article were defrayed in part by the payment of page charges. The article must therefore be hereby marked "advertisement" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

ergometer (2). Before initiating the training period, the subjects carried out three different one-legged performance tests with each leg as described in *One-Legged Performance Tests*. Before training, there was no difference in performance between the legs, and the leg to be trained was selected randomly.

The training lasted for 8 wk, and the frequency of training increased from 3 days/wk in the first week to 5 days/wk in the fifth week. The subjects performed three sets of intermittent exercise consisting of 2×30 -s and 3×1 -min bouts interspersed with 2-min rest periods per session in the first 2 wk, four sets in the following 2 wk, and five sets in the last 4 wk. The exercise bouts were performed with the flywheel preset (flying start) at kicking frequencies of 80 and 60 rpm for the 30-s and 1-min exercise bouts, respectively. The resistance was set at 100–120 N for the 30-s bouts, and, to exhaust the subjects at the end of the 1-min bouts, a resistance of 50–100 N was selected. Each training session was preceded by a 10-min warm-up period at 10 W.

One-Legged Performance Tests

Before and at the end of the training period the subjects performed three different performance tests with both the untrained and trained leg. Each test was performed on a separate day and was preceded by a 10-min warm-up period at an intensity of 10 W.

Maximal kicking. In the supine position, the subjects exercised maximally for 30 s, with flying start at 80 rpm and resistance of 120 N. Kicking frequency was recorded continuously during the test. Peak power (the highest power output in a 5-s interval) and mean power (average power during the 30-s period) were determined.

Endurance test. In the supine position, subjects performed one-legged knee-extensor exercise at a power output of 73.5 (55.0–93.5) W to exhaustion. The work rates were selected individually based on a number of preexperiments to obtain an exercise time of ~3 min.

Maximal isometric contraction. In the supine position, the subjects performed three one-legged knee-extensor maximal isometric contractions separated by 1-min rest periods, and peak isometric force was determined.

Muscle Sampling

For three of the subjects, muscle biopsies were obtained from the experimental leg before and 48 h after the last training session. For the other four subjects, muscle biopsies were obtained both from the trained and the untrained leg 48 h after completion of the training period. The subjects reported to the laboratory, and after 30 min of rest in the supine position three to four needle muscle biopsies were obtained from the same site of the trained and the untrained muscle. About 250 mg were used for sarcolemmal giant vesicle formation; one part of the biopsies was prepared for histochemical analyses, and the remaining portion (10–50 mg) was frozen in liquid N₂.

Muscle Analyses

Sarcolemmal giant vesicle transport. The rate of lactate transport was determined in sarcolemmal giant vesicles produced by collagenase treatment of muscle biopsies as previously described (14, 17). Vesicles were inspected by phase contrast microscopy, and the diameters measured ranged from 2 to 25 μ m with a median diameter of 4.4 μ m, which is similar to previous findings for human skeletal muscle (27). Average vesicle diameter distributions were

determined for the untrained and the trained muscles, and there was no difference between the distributions. The lactate efflux was determined during zero-trans conditions with 30 mM lactate. Briefly, vesicles were preloaded with lactate and radiolabeled lactate. The efflux was measured upon transferring the vesicles to a lactate-free medium. Fourteen samples were collected from the medium within the first 10 minutes, and the initial rate of efflux was calculated from a curve fit, as previously described (27). The carrier-mediated lactate transport was estimated by subtracting the flux, which previously has been found to occur via simple diffusion, from the total lactate transport (27) assuming that the diffusion component was similar in the trained and untrained muscle.

MCT isoform determination. In four of the subjects, MCT isoforms were determined in samples from untrained and trained legs. Total crude homogenate samples were prepared and loaded in equal amounts of protein in each lane on the same gel. Subsequently, sodium dodecyl sulfate-polyacrylamide gel electrophoresis was run, and proteins were transferred to an Immobilon P-polyvinylidene difluoride membrane (Millipore IPVH 00010) as previously described (28). Polyclonal antibodies were raised to human MCT1 and MCT4 in rabbits using the protocol previously described (30). These peptides contained the COOH-terminal 16 residues of MCT1 (12) and MCT4 (31), respectively, supplemented with an NH₂-terminal cysteine for coupling to keyhole limpet hemocyanin. A horseradish peroxidase-labeled goat anti-rabbit antibody was used as secondary antibody, and antibody-antigen complexes were visualized by the enhanced chemiluminescence detection method. Densitometric scanning (SigmaGel) of the blots was performed for quantification of the content of MCT1 and MCT4 in the muscle samples.

Muscle enzyme and buffer capacity. A part of the muscle samples was freeze-dried and analyzed fluorometrically for citrate synthase, L-3-hydroxyacyl-CoA-dehydrogenase (HAD), phosphofructokinase, total lactate dehydrogenase (LDH), and LDH₃₊₄₊₅ activities (18). Muscle pH was measured by a small glass electrode (Radiometer GK2801) after homogenization of freeze-dried samples in a nonbuffered solution containing 145 mM KCl, 10 mM NaCl, and 5 mM sodium fluoride (19, 26). After having adjusted pH of the sample to 7.1 with 0.01 M NaOH, the sample was titrated to pH 6.0 by serial additions of 0.01 M HCl followed by titration back to pH 7.1 by serial additions of 0.01 M NaOH. The pH was measured after each addition. The non-HCO₃⁻ physiochemical buffer capacity was determined from the number of moles of H⁺ required to change pH from 7.1 to 6.5 and was expressed as millimoles H⁺ per kilogram wet weight per pH (19, 26).

Histochemical analysis. The part of the muscle biopsies that was used for histochemical analyses was mounted in embedding medium, frozen in isopentane precooled in liquid N₂, and stored at -80°C until analyzed. Serial transverse sections were cut in a cryostat and either stained to characterize the fibers as type I, type IIA, and type IIB (7) or to visualize the capillaries (1). Fiber areas, fiber types, and capillary density were determined using a COMMAS image scanner (SBsysCOMFAS; Scan Beam, Hadsund, Denmark).

In Vivo Experiments

Protocols. To determine lactate and H⁺ release during intense exercise, five of the subjects carried out one out of two one-legged knee-extensor protocols 2 days after the last training session. In both protocols the subjects arrived at the laboratory after a light breakfast, and after 15 min of supine rest catheters were inserted in a femoral artery and vein of

each leg. After another 30 min of rest in the supine position, the subjects performed one-legged (the leg was randomly selected) exercise at 10 W for 10 min as warm-up and then rested for another 10 min.

Two of the subjects performed continuous incremental (20–70 W) one-legged exercise until exhaustion, and after 1 h of rest the protocol was repeated with the other leg. The average exercise duration was 698 (690 and 705) and 793 (715 and 870) s for the untrained and the trained leg, respectively. The other three subjects performed one-legged knee-extensor exercise at a constant power output until exhaustion with both the untrained [78.5 (range: 52.0–93.9) W] and trained [78.0 (range: 54.1–91.5) W] leg separated by 105 min. The duration of the exercise for the untrained and trained leg was 165 (130–217) and 260 (203–301) s, respectively.

Arterial and venous blood samples were drawn simultaneously from the exercising leg at the end of exercise. The samples were instantly placed on ice, a portion of each sample was centrifuged rapidly (<15 s), and plasma was collected. Femoral venous blood flow during exercise was measured by the thermodilution technique (3) immediately before blood sampling. Muscle biopsies were obtained from the middle portion of the vastus lateralis muscle at exhaustion, and the samples were frozen rapidly in liquid N₂.

BLOOD ANALYSIS. Oxygen saturation and hemoglobin (Hb) concentration of the blood samples were determined using an OSM-3 (Radiometer, Copenhagen, Denmark). Blood pH and partial pressures of oxygen and carbon dioxide (P_{CO₂}) were measured with the Astrup technique, and from these the actual base excess (ABE) was calculated (ABL 30; Radiometer; see Ref. 33). The ABE values were adjusted (ABE_{adj}) by taking into account saturation of Hb as described by Siggaard-Andersen (33). Whole blood and plasma lactate concentrations were measured on a YSI analyzer (Yellow Springs Instruments, Yellow Springs, OH).

MUSCLE ANALYSIS. The water content of the muscle samples was determined by weighing the samples before and after freeze-drying. Muscle lactate concentrations of freeze-dried samples were determined fluorometrically (18). Muscle pH was measured as described in *Muscle enzyme and buffer capacity*.

Calculations. Lactate and H⁺ release across the thigh was determined by multiplying blood flow by the venous-arterial concentration difference.

The cellular lactate concentration (mmol/l cell water) was calculated as $([Lac]_m - 0.15 \cdot [Lac]_{vpl}) / (H_2O_m - 0.15)$, where $[Lac]_m$ is the wet weight lactate concentration in the biopsy, $[Lac]_{vpl}$ is the venous plasma lactate concentration corrected for an assumed 6% protein content in venous plasma, H_2O_m is the fractional water content in the biopsy, and 0.15 is the assumed fractional interstitial volume of total muscle water content during exercise (34). The venous plasma lactate concentration was considered to be the best estimate of the interstitial lactate concentration. The measured H⁺ concentration in the homogenized biopsy was assumed to represent the cellular H⁺ concentration. Lactate and H⁺ gradients between the cellular space and the interstitium were estimated as the differences between the muscle cellular concentrations and the venous plasma concentrations.

Statistics

Data are presented as means $\pm$ SE. The results from the untrained and trained leg were compared using the Wilcoxon ranking test for paired data. A significance level of 0.05 was chosen.

Table 1. Peak and mean power during 30-s maximal one-legged knee-extensor kicking, work performed during intense exhaustive knee-extensor exercise, and peak knee-extensor isometric force for UT and T leg

	UT	T
30 s Maximum		
Peak power, W	185 $\pm$ 10	214 $\pm$ 12*
Mean power, W	164 $\pm$ 6	189 $\pm$ 10*
Endurance test, kJ	9.1 $\pm$ 1.5	21.1 $\pm$ 6.1*
Peak isometric force, N	566 $\pm$ 36	657 $\pm$ 46*

Values are means $\pm$ SE; $n = 7$ subjects. UT, untrained; T, trained. *Significantly different from untrained leg, $P < 0.05$.

RESULTS

Performance

For the trained leg, peak and mean power during the 30-s maximal knee extensor exercise test was 16 and 15%, respectively, higher ($P < 0.05$) than for the untrained leg (Table 1). In addition, the work performed during the endurance exercise test was 132% greater ($P < 0.05$), and peak isometric force was 16% higher ($P < 0.05$) for the trained than for the untrained leg.

Lactate Transport and MCT Expression

The rate of lactate transport in sarcolemmal giant vesicles obtained from the trained muscle was higher ($P < 0.05$) than in vesicles from the untrained muscle (64.3 ± 4.2 vs. 57.2 ± 2.6 pmol $\cdot$ cm⁻² $\cdot$ s⁻¹).

Representative Western blots showing the expression of MCT1 and MCT4 in the untrained and trained muscle of two subjects are given in Fig. 1. The average contents of MCT1 and MCT4 protein in the trained muscle were 70 ± 32 and $33 \pm 10\%$ higher ($P < 0.05$), respectively, than in the untrained muscle.

Muscle Buffer Capacity and Enzymes

The non-HCO₃⁻ physiochemical buffer capacity was the same for the trained and untrained muscle (45.9 ± 4.1 vs. 44.9 ± 4.8 mmol $\cdot$ kg wet wt⁻¹ $\cdot$ pH⁻¹). The activity of HAD in the trained muscle was 20% higher ($P < 0.05$) than in the untrained muscle, whereas there were no significant differences in the activity of citrate synthase, phosphofructokinase, or LDH (Table 2).

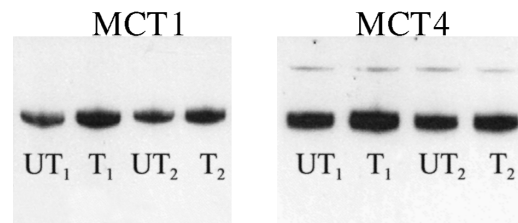


Fig. 1. Representative Western blots showing the content of monocarboxylate transporters MCT1 and MCT4 in homogenized muscle samples made from untrained (UT) and trained (T) muscle of two subjects. Subscript nos. 1 and 2 indicate subjects 1 and 2, respectively.

Table 2. PFK, HAD, CS, and LDH activities in UT and T vastus lateralis muscle as well as MCT1 and MCT4 expression in relation to UT muscle

	PFK	HAD	CS	LDH total	LDH 1+2	LDH 3+4+5	MCT1, %	MCT4, %
UT	106 ± 6	41 ± 3	38 ± 2	556 ± 60	185 ± 26	371 ± 35	100	100
T	121 ± 15	48 ± 3*	45 ± 1	598 ± 107	181 ± 26	417 ± 83	170 ± 32	133 ± 10

Values are means ± SE; $n = 7$ subjects, except for MCT1 and MCT4, where $n = 4$. Phosphofructokinase (PFK), β -hydroxyacyl-CoA-dehydrogenase (HAD), citrate synthase (CS), and lactate dehydrogenase (LDH) activities ($\text{mmol} \cdot \text{kg wet wt}^{-1} \cdot \text{min}^{-1}$) in untrained (UT) and trained (T) vastus lateralis muscle as well as monocarboxylate transporter MCT1 and MCT4 expression in relation to UT muscle. *Significantly different from untrained muscle, $P < 0.05$.

Fiber Type Distribution and Capillarization

There was no difference in the relative distribution of the three main fiber types in the trained and untrained muscle. The fiber area of each fiber type and the number of capillaries were also the same in the two muscles (Table 3).

Lactate Release During Exercise

Femoral arterial and venous blood and plasma lactate concentrations were higher ($n = 5$; $P < 0.05$) in the trained than in the untrained leg at the end of exercise (Fig. 2). The femoral venous-arterial lactate difference at the end of exercise was similar in trained compared to untrained muscle (2.33 ± 0.07 vs. 1.97 ± 0.21 mmol/l, $n = 5$). Leg blood flow was 5.64 ± 0.45 and 5.86 ± 0.31 l/min at the end of exercise in the untrained and trained leg, respectively. No difference in lactate release between the trained (13.7 ± 0.9 mmol/min) and untrained (11.7 ± 1.7 mmol/min) leg was observed.

Muscle lactate concentration at exhaustion was lower ($P < 0.05$) in the trained than in the untrained muscle (13.3 ± 2.6 vs. 17.7 ± 2.9 mmol/kg wet wt). Similarly, the estimated intracellular-to-interstitial gradient of lactate at the point of exhaustion was lower ($P < 0.05$) in the trained than in the untrained muscle (4.8 ± 3.3 vs. 17.4 ± 4.3 mmol/l).

H⁺ Release and Acid-Base Balance During Exercise

The femoral arterial and venous actual base excess corrected for degree of Hb saturation (ABE_{adj}) was lower ($P < 0.05$) in the trained than in the untrained

leg at the end of exercise (Fig. 2A), and the arterial-venous difference in ABE_{adj} was similar in trained (4.93 ± 0.85 mmol/l) and untrained (4.56 ± 0.31 mmol/l) legs. The estimated H⁺ release in the trained and untrained legs was the same at the end of exercise (29.3 ± 5.3 vs. 26.1 ± 3.5 mmol/min).

Muscle pH at exhaustion was the same in the trained and untrained muscle (6.97 ± 0.04 vs. 6.86 ± 0.06). The estimated intracellular-to-interstitial H⁺ gradient was lower ($P < 0.05$) in the trained than in the untrained muscle (85.1 ± 19.9 vs. 129.6 ± 18.3 nmol/l).

Arterial and venous pH and PCO_2 were the same in the two legs at exhaustion, whereas the arterial blood HCO_3^- concentration was lower ($P < 0.05$) in the trained than in the untrained leg at the end of exercise (Fig. 3).

DISCUSSION

The main findings of the present study were that high-intensity training induced an increase in the sarcolemmal lactate/H⁺ transport capacity as well as an enhanced content of MCT1 and MCT4 protein in human skeletal muscle. These changes were associated with similar release rates of lactate and H⁺ during intense exercise before and after training, although the intracellular-to-interstitial gradients of lactate and H⁺ were lower after training. Combined, these results suggest that the muscle lactate/H⁺ transporters can be altered by intense training and that they play an important role in the regulation of lactate and pH in human skeletal muscle.

Training has been reported to enhance the lactate transport capacity of rat skeletal muscle (24, 28). Furthermore, the observation that athletes have a higher capacity to transport lactate than untrained and less trained subjects (27) indicates that training also can affect lactate/H⁺ transport in humans. The present findings demonstrate for the first time that high-intensity training can elevate the rate of sarcolemmal lactate/H⁺ transport in human skeletal muscle. It cannot be excluded that a difference in lactate/H⁺ transport capacity existed between the two legs already before the training was initiated, because for four of the subjects biopsies were obtained from both legs at the end of the training period. However, this possibility seems unlikely, since the performance ability was similar for the two legs before the training period for these

Table 3. Fiber type distribution, fiber area and capillarization of UT and T vastus lateralis muscle

	UT	T
Fiber type distribution, %		
I	64.4 ± 4.2	63.3 ± 8.8
IIA	22.1 ± 3.7	27.0 ± 3.8
IIB	13.2 ± 2.9	9.1 ± 3.4
Fiber area, μm^2		
I	5,571 ± 788	5,914 ± 675
IIA	6,545 ± 1,022	6,923 ± 1,274
IIB	6,118 ± 1,370	6,605 ± 1,407
Capillarization		
number/mm ²	393 ± 16.5	393 ± 22.7
number/fiber	5.4 ± 0.6	5.5 ± 0.4

Values are means ± SE; $n = 7$ subjects.

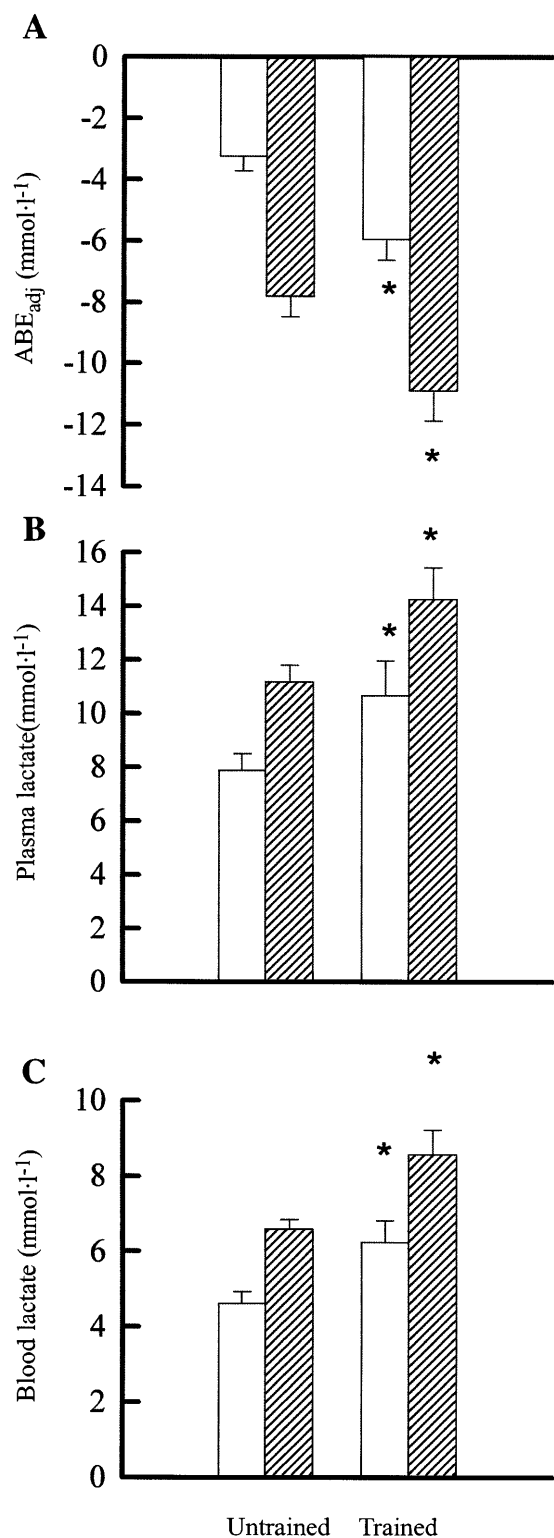


Fig. 2. Femoral arterial (open bars) and venous (hatched bars) blood actual base excess (A) and plasma (B) and blood (C) lactate concentration at the end of intense knee-extensor exercise in untrained and trained leg. Values are means $\pm$ SE; $n = 5$ subjects. ABE_{adj}, adjusted actual base excess. *Significantly different from untrained leg, $P < 0.05$.

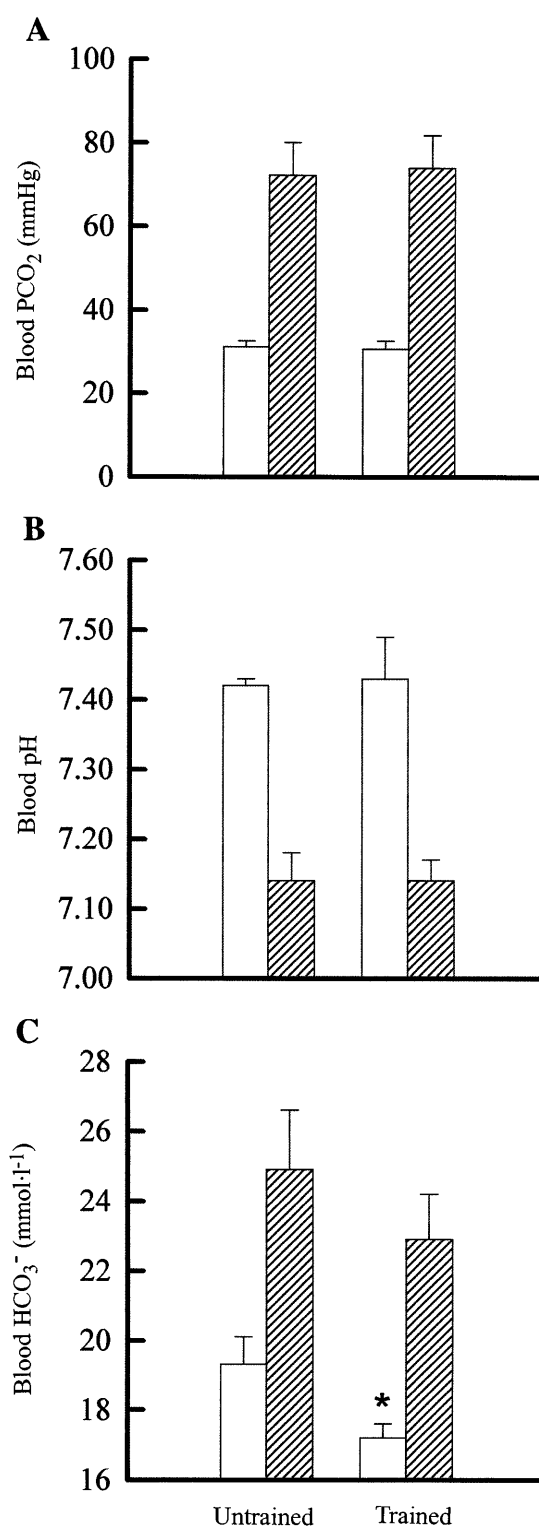


Fig. 3. Femoral arterial (open bars) and venous (hatched bars) blood PCO₂ (A), pH (B), and HCO₃⁻ (C) at the end of intense knee-extensor exercise in untrained and trained leg; $n = 5$ subjects. Values are means $\pm$ SE. *Significantly different from untrained leg, $P < 0.05$.

subjects, and the rate of sarcolemmal lactate transport was enhanced in all three subjects in whom biopsies were obtained from the same leg before and after training.

The higher content of MCT1 and MCT4 protein in the trained muscle can in part explain the observed increase in the sarcolemmal lactate/ H^+ transport. However, in addition to MCT1 and MCT4, mRNA for other MCT isoforms have been found in human skeletal muscle (31), and the protein quantity of these may also have been altered. In addition, the relative contributions of MCT1 and MCT4 to total lactate/ H^+ transport are not known. For these reasons, it is not to be expected that the relative changes in MCT1 and MCT4 are reflected in similar relative changes in vesicular lactate transport. In addition, it is likely that free diffusion of lactate at a given muscle lactate gradient is unaltered by training. Therefore, the relative change in carrier-mediated lactate transport was presumably larger than the observed percentage change in vesicular lactate transport.

Determinations of MCT1 and MCT4 in various rat muscles have revealed that MCT4 exists in both oxidative and glycolytic muscles (36), whereas MCT1 predominantly exists in oxidative muscles (23). Therefore, it could be that the presently observed more marked increase in MCT1 compared with the change in MCT4 was due to a smaller basal content of MCT1 than of MCT4 in the muscle, since the relative contribution of fast-twitch fibers was ~35%. The training-induced increase in MCT1 and MCT4 was not caused by a general elevation in muscle proteins, since the MCT content is expressed per milligram of total muscle protein. On the other hand, the lactate/ H^+ transporter proteins were not the only membrane proteins that were affected, as evidenced by the number of Na^+ - K^+ pumps in the plasma membrane, which were 13% higher in the trained than in the untrained muscle in the present study (unpublished data). Furthermore, the higher HAD activity in the trained than in the untrained muscle shows that the training also elicited adaptations in mitochondrial proteins, whereas no significant changes were observed in other muscle enzymes.

To evaluate the importance of the training-induced improvement in lactate/ H^+ transport, release rates of lactate and H^+ were examined during short-lasting, exhaustive exercise with the trained and untrained leg in five of the subjects. The finding of similar release rates of lactate and H^+ in the two legs at the point of exhaustion, despite a lower intracellular-to-interstitial gradient of lactate and H^+ in the trained than in the untrained muscle (Fig. 4), may be explained by the enhanced sarcolemmal lactate/ H^+ transport capacity in the trained muscle.

An additional reason for the need of a lower driving force to obtain a given transmembrane flux of H^+ in the trained muscle could be that the capacity of the Na^+ / H^+ exchange system also increased in response to the high-intensity training, as has been demonstrated in

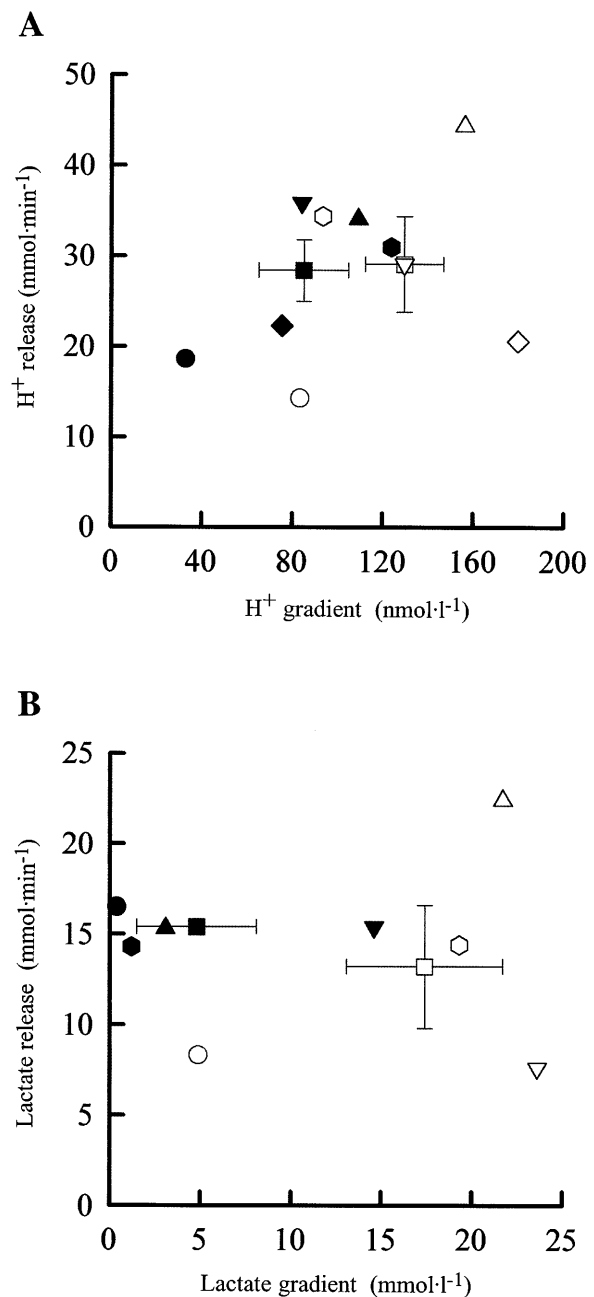


Fig. 4. Release of H^+ related to intracellular-to-interstitial gradient of H^+ (A) and release of lactate related to intracellular-to-interstitial gradient of lactate (B). Open and filled symbols indicate untrained and trained muscles, respectively. Open and filled squares denote average values $\pm$ SE, whereas other symbols represent individual data.

rats (16). The training solely affected muscle pH regulation via changes in membrane transport systems, since the training had no influence on the physiochemical buffer capacity, which is in accordance with previous reports (20, 25).

In summary, the present data show that high-intensity training can enhance the sarcolemmal lactate/ H^+ transport capacity as well as the contents of MCT1 and MCT4 protein in human skeletal muscle. These changes were associated with reduced transmem-

brane gradients of lactate and H^+ in the trained leg for a given release rate of lactate and H^+ . This suggests that the training-induced changes in lactate/ H^+ transport are of importance for the regulation of muscle lactate and pH in humans.

We thank B. Saltin for providing medical assistance. A. Honig, M. Vannby, I. Kring, and W. Taagerup provided skillful technical assistance.

This research was supported by the Danish National Research Foundation (J.nr. 504–14). In addition, support was obtained from Team Danmark and Idrættens Forskningsråd.

Address for reprint requests: H. Pilegaard, August Krogh Institute, Univ. of Copenhagen, Universitetsparken 13, 2100 Copenhagen, Denmark.

Received 16 June 1998; accepted in final form 8 October 1998.

REFERENCES

- Andersen, P. Capillary density in skeletal muscle in man. *Acta Physiol. Scand.* 95: 203–205, 1975.
- Andersen, P., R. P. Adams, G. Sjøgaard, A. Thorboe, and B. Saltin. Dynamic knee extension as model for study of isolated exercising muscle in humans. *J. Appl. Physiol.* 59: 1647–1653, 1985.
- Andersen, P., and B. Saltin. Maximal perfusion of skeletal muscle in man. *J. Physiol. (Lond.)* 366: 233–249, 1985.
- Baker, S. K., K. J. A. McCullagh, and A. Bonen. Training intensity-dependent and tissue-specific increases in lactate uptake and MCT-1 in heart and muscle. *J. Appl. Physiol.* 84: 987–994, 1998.
- Bergström, J. Muscle electrolytes in men. *Scand. J. Clin. Lab. Invest. Suppl.* 68: 11–13, 1962.
- Bonen, A., K. J. A. McCullagh, C. T. Putman, E. Hultman, N. L. Jones, and G. J. F. Heigenhauser. Short-term training increases human muscle MCT1 and femoral venous lactate in relation to muscle lactate. *Am. J. Physiol.* 274 (Endocrinol. Metab. 37): E102–E107, 1998.
- Brooke, M. H., and K. K. Kaiser. Three myosin adenosine triphosphatase systems: the nature of their pH lability and sulphhydryl dependence. *J. Histochem. Cytochem.* 18: 670–672, 1970.
- Favero, T. G., A. C. Zable, D. Colter, and J. J. Abramson. Lactate inhibits Ca^{2+} -activated Ca^{2+} -channel activity from skeletal muscle sarcoplasmic reticulum. *J. Appl. Physiol.* 82: 447–452, 1997.
- Fitts, R. H. Cellular, molecular, and metabolic basis of muscle fatigue. *Handbook of Physiology. Exercise: Regulation and Integration of Multiple Systems*. Bethesda, MD: Am. Physiol. Soc., 1996, sect. 12, chap. 23, p. 1036–1074.
- Garcia, C. K., M. S. Brown, R. K. Pathak, and J. L. Goldstein. cDNA cloning of MCT2, a second monocarboxylate transporter expressed in different cells than MCT1. *J. Biol. Chem.* 270: 1843–1849, 1995.
- Jackson, C. K., X. Li, J. Luna, and U. Francke. cDNA cloning of the human monocarboxylate transporter 1 and chromosomal localization of the SLC16A1 locus to 1p13.2–p12. *Genomics* 23: 500–503, 1994.
- Jackson, V. N., N. T. Price, and A. P. Halestrap. cDNA cloning of MCT1, a monocarboxylate transporter from rat skeletal muscle. *Biochim. Biophys. Acta* 1238: 193–196, 1995.
- Juel, C. Intracellular pH-recovery and lactate efflux in mouse soleus muscles stimulated in vitro: the involvement of sodium/proton exchange and a lactate carrier. *Acta Physiol. Scand.* 132: 363–371, 1988.
- Juel, C. Muscle lactate transport studied in sarcolemmal giant vesicles. *Biochim. Biophys. Acta* 1065: 15–20, 1991.
- Juel, C. Regulation of cellular pH in skeletal muscle fibre types, studied with sarcolemmal giant vesicles obtained from rat muscles. *Biochim. Biophys. Acta* 1265: 127–132, 1995.
- Juel, C. Skeletal muscle Na^+/H^+ exchange in rats: pH dependency and the effect of training. *Acta Physiol. Scand.* 164: 135–140, 1998.
- Juel, C., S. Kristiansen, H. Pilegaard, J. Wojtaszewski, and E. A. Richter. Kinetics of lactate transport in sarcolemmal giant vesicles obtained from human skeletal muscle. *J. Appl. Physiol.* 76: 1031–1036, 1994.
- Lowry, O., and J. V. Passonneau. *A Flexible System on Enzymatic Analysis*. New York: Academic, 1972.
- Mannion, A. F., P. M. Jakeman, and P. L. T. Willan. Determination of human skeletal muscle buffer value by homogenate technique: methods of measurement. *J. Appl. Physiol.* 75: 1412–1418, 1993.
- Mannion, A. F., P. M. Jakeman, and P. L. T. Willan. Effects of isokinetic training of the knee extensors on high-intensity exercise performance and skeletal muscle buffering. *Eur. J. Appl. Physiol.* 68: 356–361, 1994.
- McCullagh, K. J. A., C. Juel, M. O'Brien, and A. Bonen. Chronic muscle stimulation increases lactate transport in rat skeletal muscle. *Mol. Cell. Biochem.* 156: 51–57, 1996.
- McCullagh, K. J. A., R. C. Poole, A. P. Halestrap, M. O'Brien, and A. Bonen. Role of the lactate transporter (MCT1) in skeletal muscles. *Am. J. Physiol.* 271 (Endocrinol. Metab. 34): E143–E150, 1996.
- McCullagh, K. J. A., R. C. Poole, A. P. Halestrap, K. F. Tipton, M. O'Brien, and A. Bonen. Chronic electrical stimulation increases MCT1 and lactate uptake in red and white skeletal muscle. *Am. J. Physiol.* 273 (Endocrinol. Metab. 36): E239–E246, 1997.
- McDermott, J. C., and A. Bonen. Endurance training increases skeletal muscle lactate transport. *Acta Physiol. Scand.* 147: 323–327, 1993.
- Nevill, M. E., L. H. Boobis, S. Brooks, and C. Williams. Effect of training on muscle metabolism during sprinting. *J. Appl. Physiol.* 67: 2376–2382, 1989.
- Parkhouse, W. S., and D. C. McKenzie. Possible contribution of skeletal muscle buffers to enhanced anaerobic performance: a brief review. *Med. Sci. Sports Exerc.* 16: 328–338, 1984.
- Pilegaard, H., J. Bangsbo, E. A. Richter, and C. Juel. Lactate transport studied in sarcolemmal giant vesicles from human muscle biopsies: relation to training status. *J. Appl. Physiol.* 77: 1858–1862, 1994.
- Pilegaard, H., C. Juel, and F. Wibrand. Lactate transport studied in sarcolemmal giant vesicles: effect of training. *Am. J. Physiol.* 264 (Endocrinol. Metab. 27): E156–E160, 1993.
- Ploug, T., B. Stallknecht, O. Pedersen, B. Kahn, T. Ohkuwa, J. Winten, and H. Galbo. The effect of endurance training on glucose transport capacity and expression of glucose transporters in rat skeletal muscle. *Am. J. Physiol.* 259 (Endocrinol. Metab. 22): E778–E786, 1990.
- Poole, R. C., C. E. Sanson, and A. P. Halestrap. Studies of the membrane topology of the rat erythrocyte H^+ /lactate cotransporter (MCT1). *Biochem. J.* 320: 817–824, 1996.
- Price, N. T., V. N. Jackson, and A. P. Halestrap. Cloning and sequencing of four new mammalian monocarboxylate transporter (MCT) homologues confirms the existence of a transporter family with an ancient past. *Biochem. J.* 329: 321–328, 1998.
- Roth, D. A., and G. A. Brooks. Lactate transport is mediated by a membrane-bound carrier in rat skeletal muscle sarcolemmal vesicles. *Arch. Biochem. Biophys.* 279: 377–385, 1990.
- Siggaard-Andersen, O. *The Acid-Base Status of the Blood*. Copenhagen, Denmark: Munksgaard, 1963.
- Sjøgaard, G., R. P. Adams, and B. Saltin. Water and ion shifts in skeletal muscle of humans with intense dynamic knee extension. *Am. J. Physiol.* 248 (Regulatory Integrative Comp. Physiol. 17): R190–R196, 1985.
- Watt, P. W., A. MacLennan, H. S. Hundal, C. M. Kuret, and M. J. Rennie. L-Lactate transport in perfused rat skeletal muscle: kinetic characteristics and sensitivity to pH and transport inhibitors. *Biochim. Biophys. Acta* 944: 213–222, 1988.
- Wilson, M. C., V. N. Jackson, C. Heddle, N. T. Price, H. Pilegaard, C. Juel, A. Bonen, I. Montgomery, O. F. Hutter, and A. P. Halestrap. Lactic acid efflux from white skeletal muscle is catalysed by the monocarboxylate transporter isoform MCT3. *J. Biol. Chem.* 273: 15920–15926, 1998.